

Reinforcement Learning Explained

AI Learning Hub - Comprehensive Course Guide

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RL Foundations

Reinforcement Learning trains agents to make sequential decisions by maximizing cumulative rewards through interaction with an environment.

Core Components:

- Agent: The learner and decision maker.
- Environment: The world the agent interacts with.
- State (s): Current situation of the agent.
- Action (a): Choices available to the agent.
- Reward (r): Feedback signal from the environment.
- Policy (π): Strategy mapping states to actions.
- Value Function $V(s)$: Expected cumulative reward from state s.
- Q-Function $Q(s,a)$: Expected cumulative reward from state s taking action a.

The Bellman Equation: $V(s) = \max_a [R(s,a) + \gamma * V(s')]$

Q-Learning and Deep Q-Networks

Q-Learning is a model-free algorithm that learns action-value function $Q(s,a)$.

Update rule: $Q(s,a) = Q(s,a) + \alpha * [r + \gamma * \max_{a'} Q(s',a') - Q(s,a)]$

Deep Q-Networks (DQN) use neural networks to approximate Q-values.

Key innovations:

- Experience Replay: Store transitions in buffer, sample randomly to break correlation between consecutive samples.
- Target Network: Separate network for computing target values, updated periodically to stabilize training.
- Double DQN: Addresses overestimation by using online network for action selection and target network for evaluation.

Policy Gradient and Actor-Critic Methods

Policy Gradient methods directly optimize the policy.

REINFORCE algorithm: Update policy parameters in direction of gradient of expected reward. High variance but unbiased.

Actor-Critic Methods combine value-based and policy-based approaches:

- Actor: Learns the policy (which action to take).
- Critic: Learns the value function (how good the state is).
- A2C (Advantage Actor-Critic): Uses advantage function $A(s,a) = Q(s,a) - V(s)$.
- PPO (Proximal Policy Optimization): Clips policy updates for stability.

Applications: Game playing (AlphaGo, Atari), robotics, autonomous driving.